

Quantum harmonic oscillator

$$L = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} m \omega^2 x^2$$

Legendre transform: $p = \frac{\partial L}{\partial \dot{x}} = m \dot{x}$

$$\begin{aligned} H &= p \dot{x} - L = m \dot{x}^2 - \frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \omega^2 x^2 \\ &= \frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \omega^2 x^2 \\ &= \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2 \end{aligned}$$

Canonical quantization: promote to operators, require $[x, p] = i$

$$\begin{aligned} H &= \frac{1}{2} (\hat{p}^2 + \hat{x}^2) = \frac{1}{2} (\underbrace{\hat{x} - i\hat{p}}_{a^\dagger}) (\underbrace{\hat{x} + i\hat{p}}_a) - \frac{1}{2} i [x, p] \\ &= a^\dagger a + \frac{1}{2} \end{aligned}$$

Look at $[H, a]$, $[a, a^\dagger]$, see

$$a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

$$a |n\rangle = \sqrt{n} |n-1\rangle$$

$$a^\dagger |0\rangle = |1\rangle$$

$$H |n\rangle = \omega \left(n + \frac{1}{2}\right) |n\rangle$$

Classical field to quantum field

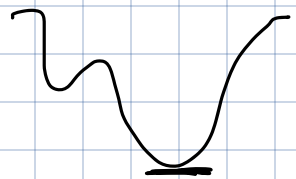
Scalar function $\varphi(t, x)$

Special relativity: $x^\mu = (t, \vec{x})$

$$\partial^\mu = \left(\frac{\partial}{\partial t}, \vec{\nabla} \right)$$

$$L = \int d^4x \mathcal{L}(\varphi, \partial_\mu \varphi)$$

can only construct $\partial_\mu \varphi \partial^\mu \varphi$ + higher powers of $(\partial \varphi)^2$

V :  $V(\varphi) = c \varphi^2 + \text{higher order}$

$$\mathcal{L} = \frac{1}{2} \partial_\mu \varphi \partial^\mu \varphi - \frac{1}{2} m^2 \varphi^2 + \text{higher order}$$

$$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\varphi}}$$

$$H = \frac{1}{2} \dot{\varphi}^2 + \frac{1}{2} (\nabla \varphi)^2 + \frac{1}{2} m^2 \varphi^2 = \frac{1}{2} \pi^2 + \frac{1}{2} (\nabla \varphi)^2 + \frac{1}{2} m^2 \varphi^2$$

Fourier transform:

$$\varphi(\vec{x}) = \int d^3 p \tilde{\varphi}(\vec{p}) e^{i \vec{p} \cdot \vec{x}}, \text{ same for } \pi$$

$$H = \int d^3 p \left[\frac{1}{2} \tilde{\pi}^2 + (\vec{p}^2 + m^2) \tilde{\varphi}^2 \right] \quad \text{sum of harmonic oscillators!}$$

Enforce commutation rels, Define $a(p), a^\dagger(p)$.

Consider $|\psi\rangle = a^\dagger(p)|0\rangle$

$$(\hat{H}, \hat{P}^i) |\psi\rangle = (E, \vec{p}) |\psi\rangle$$

$a^\dagger(p)|0\rangle$ creates part. w/ 4-mom p^μ

$\varphi(\vec{x})|0\rangle$ creates part at position $\vec{x}$

$\left. \begin{array}{l} \varphi \text{ is a quark} \\ \text{field.} \\ \text{transforms diff.} \\ \text{under Lorentz} \\ \text{+ translations} \end{array} \right\}$

Parton distrib functions

$$\sigma(x p \rightarrow y z) = \sum_i \int_0^1 d\bar{\xi} f_i(\bar{\xi}) \sigma(x p_i \rightarrow y z) \quad \text{"factorization"}$$

$$p_i = \bar{\xi} p$$

$\hookrightarrow$ universal!

$\hookrightarrow$ perturbative

$$f_q(\xi) = \sum_x \int d\pi_x |\langle x | \psi(0) | P \rangle|^2 \delta(\xi n \cdot P - n \cdot p)$$

↳ allows transverse fluctuations
only n components need
match

Note $p + p_x = P$. Write

$$\delta(\xi n \cdot P - n \cdot p) = \int_{-\infty}^{\infty} dt e^{-itn \cdot (\xi P - P + p_x)}$$

Notice this is like momentum operator

$$e^{-itn \cdot (\xi P - P + p_x)} \langle P | \psi^\dagger | x \rangle$$

$$= e^{-i\xi t n \cdot P} \langle P | e^{in \cdot \vec{p} t} \psi^\dagger(0) e^{-in \cdot \vec{p} t} | x \rangle$$

$$= e^{-i\xi t n \cdot P} \langle P | \psi^\dagger(tn) | x \rangle$$

Now no more x dependence... use $\int d\pi_x |x\rangle \langle x| = 1$

$$\int_{-\infty}^{\infty} dt e^{-it\xi n \cdot P} \langle P | \bar{\psi}(tn) \gamma^0 \psi(0) | P \rangle$$

write $\gamma^0 \psi \approx \not{n} \psi$

Gauge invariance... add Wilson line W_n

$$f_q(\xi) = \int_{-\infty}^{\infty} dt e^{-it\xi n \cdot P} \langle P | \bar{\psi}(tn) \not{n} W_n \psi(0) | P \rangle$$

fragmentation functions... we are interested in final state of

$J/\psi + X$. keep that sum over X . Initial state is vacuum

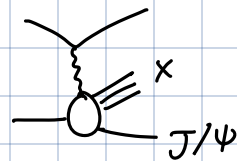
instead of proton... care about fragmentation happening at all,

not just in proton.

↳ we want to
compute this

$$F_q(z) = \int db_- \sum_x e^{-ib_- n \cdot P/z} \not{n} \langle 0 | W \psi(tn) | J/\psi, x \rangle \langle J/\psi, x | \bar{\psi} W(0) | 0 \rangle$$

Focus on the one matrix element



$$\langle 0 | W \Psi(t_n) | J/\psi, x \rangle \approx \langle 0 | W \Psi(t_n) e^{-i \int d^4x \mathcal{H}_{int}} | c\bar{c} + x \rangle$$

$$W \sim e^{ig \int dt n \cdot A(tn)}$$

$$\mathcal{H}_{int} \sim -g \bar{\Psi} \not{A} \Psi \rightarrow 1 - ig \int d^4x \bar{\Psi} \not{A} \Psi + \frac{1}{2} (\dots)^2$$

wilson line

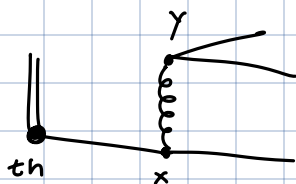


→ need this so we can get $c\bar{c}$ in final state

$$g^2 \langle 0 | \overbrace{A(x) \Psi(t_n) \bar{\Psi} \not{A} \Psi}^{\text{wilson line}} | c\bar{c} + x \rangle$$



$$g^2 \langle 0 | \Psi(t_n) \bar{\Psi}(x) \not{A}(x) \Psi(x) \bar{\Psi}(y) \not{A}(y) \Psi(y) | J/\psi + x \rangle$$



Then use Feynman rules (derive using integrals over d^4x, d^4y w/ position space contraction rules)

Look closely at

$$\int \frac{d^4k}{(2\pi)^4} \frac{d^4p}{(2\pi)^4} \frac{1}{k^2} \frac{1}{(k+p)^2} \frac{1}{p^2}$$

$$\bar{u}(k) \gamma^\mu \frac{\cancel{k} + \cancel{p}}{(k+p)^2} \bar{u}(p/2) \gamma_\mu v(p/2) \cancel{v}(p/2) \gamma_\nu u(p/2) \frac{\cancel{k} + \cancel{p}}{(k+p)^2} \gamma^\nu u(k) \times \delta(k^2)$$

$$\times \int db_- e^{ib_- n \cdot p / z} \times e^{-ib(k+p)} \times \int d^4 k$$

↳ bc Wilson line was a bit different on RHS

Then use delta fns...

$$\int db_- e^{ib_- n \cdot p / z} \times e^{-ib(k+p)} = e^{i\vec{k}_T \cdot \vec{b}_T} \delta\left(\frac{1-z}{z} p_- - k_-\right)$$

$$\delta(k^2) = \delta(2k_+ k_- - \vec{k}_T^2)$$

This lets you do k_+/k_- integrals. Left w/ TMD dependence (k_T) .